CP(M) 2026 Invigilated MiniZinc Assessment, mock paper: sample answers at about band A

Question 1.1. By induction on t.

For $t = 0$, (D) on line 55 makes `burnt[0, v]` false for every v, and in the burning process nothing is burning at turn 0.

Now suppose that $t \geq 1$ and that `burnt[t-1, v]` is true exactly when v is burning at the end of turn $t-1$. Constraint (E) on lines 58-60 is an equivalence, so `spread[t, v]` is true exactly when v or one of its neighbours is burning at the end of turn $t-1$. These are exactly the vertices that are burning after the spreading step of turn t. Constraint (F) on lines 63-65 is also an equivalence, so `burnt[t, v]` is true exactly when `spread[t, v]` is true or `burn_turn[v]` $= t$. These are the vertices burning after the spread, together with the vertex chosen at turn t, which is exactly the set of vertices burning at the end of turn t.

Because (E) and (F) are equivalences, both directions hold: the constraints force `burnt[t, v]` to be false for every vertex that is not burning, as well as true for every vertex that is. The argument also covers the turns after `turns`, at which no vertex has `burn_turn[v]` $= t$ and the fire simply spreads.

Question 1.2. No. Take K_2, with vertices 1 and 2. Here `UB` $= \mathrm{rad}(K_2) + 1 = 2$ and $b(K_2) = 2$.

- `turns` $= 1$ is impossible, since only the chosen vertex is burning at the end of turn 1.
- For `turns` $= 2$, constraints (A)-(C) on lines 44, 47 and 50 force a vertex to be chosen at turn 2. Whichever vertex was chosen at turn 1, the spread at turn 2 reaches the other one, so `spread[2, 1]` and `spread[2, 2]` are both true. The modified (H) then forbids both choices at turn 2.

So the modified model has no solution on K_2, although K_2 can be burned in two turns.

The deleted disjunct is what allows a choice when the fire has already reached every vertex. Constraints (A)-(C) require a choice at every turn $1, \ldots,$ `turns`, so without the disjunct the model fails whenever every optimal sequence has the fire reaching everything before its last choice. This does not happen on every graph. On `path-9`, for example, the sequence $(3, 7, 9)$ chooses vertex 9 before the fire has reached it, so the modified model still finds $b = 3$.

Question 2.1. `UB` $= 9$, so `RADIUS` $= 0..8$. By (A) on line 53, `y[r]` $= 1$ exactly when some vertex has radius r, so `y[0]` $=$ `y[1]` $=$ `y[2]` $=$ `y[3]` $= 1$ and `y[4]` $= \cdots =$ `y[8]` $= 0$. By (C) on line 59, `turns` $= 4$.

A vertex with radius r is chosen at turn `turns - r`, so `burn_turn[4]` $= 1$, `burn_turn[10]` $= 2$, `burn_turn[14]` $= 3$, `burn_turn[16]` $= 4$, and `burn_turn[v]` $= 0$ for every other v. The output prints `sequence = [4, 10, 14, 16]`.

For $u = 7$, $d(7, 4) = 3 \leq 3$, so `x[4, 3]` is in the sum of (E) and the sum is at least 1. In fact it is exactly 1, since $d(7, 10) = 3 > 2$, $d(7, 14) = 7 > 1$ and $d(7, 16) = 9 > 0$. For $u = 13$, $d(13, 14) = 1 \leq 1$, so `x[14, 1]` is in the sum. The other fires do not cover 13: $d(13, 4) = 9 > 3$,

$d(13, 10) = 3 > 2$ and $d(13, 16) = 3 > 0$. The four fires cover 1..7, 8..12, 13..15 and 16, so every vertex is covered exactly once.

Question 2.2. Both `turns` and `x[v, r]` are decision variables, so `(turns - r) * x[v, r]` is a product of two variables, which is not linear. In (F), when v is not chosen, `chosen` $= 0$ and the inequality on line 79 becomes `burn_turn[v] >= turns - M`. This must allow `burn_turn[v]` $= 0$, which needs $M \geq$ `turns`, and `turns` is at most `UB` by its domain on line 48.

Question 3, first part. In both models, `int: k;` must be added as a parameter.

In Appendix A, replace (A) on line 44 by

```
constraint forall(t in 1..UB)(count(v in 1..n)(burn_turn[v] = t) <= k);
```

and delete (C) on line 50. With up to k choices per turn, some turns may have fewer than k choices, or none, so the number of chosen vertices no longer tells us `turns`. Keeping (C) would wrongly require exactly `turns` chosen vertices in total. Keep (B) on line 47. Constraint (H) is still valid and can be left as it is. It could now be strengthened to `burn_turn[v] = t -> not spread[t, v]`, because a turn may have no choice at all, unlike in Question 1.2.

In Appendix B, replace (A) on line 53 by

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constraint forall(r in RADIUS)(sum(v in 1..n)(x[v, r]) <= k * y[r]);
```

and keep (B)-(F). (F) still works, because (D) still allows each vertex to be chosen at most once. In both models the output item prints the chosen vertices as one list, so I would change it to print the set of vertices chosen at each turn.

Question 3, second part. Choose vertices 3 and 8 at turn 1, vertices 12 and 15 at turn 2, and nothing at turn 3. By the end of turn 3, the fires lit at turn 1 have spread twice and cover 1..5 and 6..10, and the fires lit at turn 2 have spread once and cover 11..13 and 14..16.

Two turns are not enough. In a two-turn process, each fire lit at turn 1 has spread once by the end of turn 2, so it covers at most 3 vertices of a path, and each fire lit at turn 2 covers only itself. With at most two fires per turn, two turns burn at most $2 \cdot 3 + 2 \cdot 1 = 8 < 16$ vertices.

Comment. Every part is correct and complete, and the answers refer to specific constraints and lines. Question 1.1 is a proper induction that explains why both directions hold. Question 1.2 gives a checked counterexample and explains which part of the model causes the failure and on which graphs. The answer to Question 3 says why (C) must be deleted, and links the strengthening of (H) back to Question 1.2. This shows an understanding of why the modelling choices matter beyond describing them, as the A band requires.